

BERKELEY COUNTY AP, SC, USA

WMO: 720598

Lat: 33.183N

Lon: 80.033W

Elev: 73

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00191

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.6	28.4	7.5	8.1	31.4	12.4	10.4	36.5	18.7	63.3	16.4	58.4	2.7	30	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	95.2	77.4	92.8	76.7	90.7	76.2	80.4	89.4	79.3	87.6	78.4	86.3	5.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.8	145.0	83.2	77.2	141.8	82.7	76.6	139.2	82.2	43.7	88.5	42.9	87.7	42.0	86.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.5	12.6	11.0	DB	19.8	97.8	4.7	3.3	16.4	100.1	13.6	102.1	10.9	103.9	7.5	106.2	(4)
			WB	17.7	81.4	4.4	1.9	14.6	82.7	12.0	83.8	9.5	84.9	6.2	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.8	47.0	51.8	57.9	65.1	73.3	79.4	81.6	80.7	76.6	66.9	55.9	52.1
	DBStd	13.91	10.21	9.26	8.81	7.08	5.40	4.39	3.19	2.85	3.96	6.91	8.10	9.01
	HDD50	407	181	82	31	2	0	0	0	0	0	0	28	83
	HDD65	2052	563	378	251	87	8	0	0	0	0	62	292	411
	CDD50	6158	88	131	275	453	721	881	980	953	797	526	205	148
	CDD65	2328	4	8	30	89	264	431	515	488	347	122	19	11
	CDH74	20723	23	69	280	779	2157	4160	5166	4451	2656	849	98	35
	CDH80	8209	1	6	58	186	750	1872	2340	1842	958	189	6	1
Wind	WSAvg	4.0	4.1	4.7	4.9	5.1	4.3	3.7	3.8	3.1	3.4	3.7	3.9	3.9
Precipitation	PrecAvg	47.0	3.3	3.6	3.7	3.0	3.4	5.2	5.3	5.9	4.1	3.5	2.6	3.4
	PrecMax	58.0	7.2	8.2	7.6	6.3	6.8	9.9	10.9	10.9	10.0	13.3	7.3	7.3
	PrecMin	34.4	1.3	1.4	0.5	0.9	1.1	2.5	2.6	3.2	0.3	0.0	0.2	0.8
	PrecStd	6.9	1.8	1.7	1.6	1.3	1.5	1.9	2.0	1.9	2.0	3.0	1.8	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	81.0	85.7	88.1	95.0	99.2	99.1	96.7	91.4	88.2	81.1	78.6
		MCWB	65.6	66.0	64.7	69.3	74.9	76.8	78.1	78.9	75.8	76.2	70.1	69.0
	2%	DB	72.1	75.3	81.3	84.3	90.2	95.0	95.3	92.9	90.2	84.2	77.0	73.3
		MCWB	64.1	63.6	64.7	67.7	72.7	76.6	77.7	77.5	75.4	73.7	67.0	66.2
	5%	DB	67.5	72.1	76.8	81.8	87.6	91.3	92.8	90.7	88.1	81.7	73.1	70.5
		MCWB	60.6	60.9	62.4	66.3	71.4	75.5	77.3	77.0	75.2	71.3	65.6	65.2
	10%	DB	63.5	67.6	72.6	79.0	84.0	90.0	90.5	89.6	85.9	79.1	70.2	67.6
		MCWB	58.2	59.3	60.8	65.3	71.2	75.4	76.9	76.9	74.7	69.6	64.7	63.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.7	68.7	69.1	74.1	76.8	80.2	82.0	81.4	80.5	78.1	73.2	71.3
		MCDB	71.9	75.6	77.6	80.8	87.4	92.0	92.5	91.4	86.6	83.2	78.3	75.7
	2%	WB	65.8	66.0	67.3	72.0	75.2	78.8	80.5	80.2	78.8	76.3	70.3	68.6
		MCDB	70.7	72.5	75.4	79.8	85.3	89.3	89.9	89.0	84.9	81.7	73.3	71.7
	5%	WB	63.0	63.5	65.5	70.1	74.0	77.8	79.4	79.1	77.7	74.7	68.3	66.3
		MCDB	65.7	68.9	73.0	77.3	83.0	87.5	87.8	86.8	84.0	79.7	71.5	69.5
	10%	WB	59.2	60.7	63.4	68.2	72.9	76.9	78.4	78.2	76.4	72.7	65.2	63.4
		MCDB	63.1	66.1	70.4	74.8	81.0	86.0	86.4	85.6	82.4	77.1	69.9	66.1
Mean Daily Temperature Range	5% DB	MDBR	19.7	21.4	22.3	22.7	20.0	18.9	18.2	16.7	17.2	19.8	21.1	19.0
		MCDBR	23.2	25.6	26.3	25.7	23.4	21.1	20.5	19.1	19.8	21.1	22.9	21.8
		MCWBR	15.6	14.7	12.1	11.4	8.8	7.3	7.1	7.1	7.6	11.0	14.1	15.2
	5% WB	MCDBR	19.2	22.8	21.6	19.5	19.3	19.2	17.8	16.6	16.0	17.2	17.9	18.9
	MCWBR	16.2	15.9	11.9	10.2	8.1	7.7	7.4	7.0	7.1	10.2	13.4	14.5	
Clear-Sky Solar Irradiance	taub	0.323	0.331	0.365	0.408	0.473	0.503	0.545	0.520	0.436	0.391	0.345	0.326	
	taud	2.530	2.503	2.428	2.291	2.156	2.122	2.020	2.084	2.315	2.415	2.505	2.553	
	Ebn at Noon	284	293	289	280	261	252	241	245	263	268	273	277	
	Edh at Noon	27	30	35	42	48	50	55	51	39	32	27	25	
All-Sky Solar Radiation	RadAvg	898	1104	1460	1859	2002	2045	1964	1770	1514	1293	996	798	
	RadStd	50	113	126	125	143	142	122	101	128	145	73	66	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+362	+256

Nomenclature: See separate page